



Course:  
Assignment:  
Name:  
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CSC 360  
Part 3: Hazards  
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## Hazard Documentation

### Insertion Sort with Pipeline Hazards Highlighted

```
1  # key
2  # t0: i
3  # t1: array[i]
4  # t2: current value
5  # t3: j
6  # t4: array[j]
7  # t5: value at array[j]
8
9  .data
10  array: .word 4, 5, -1, 100, 1, 22
11  n: .word 6
12
13  .text
14  lw a1, n
15  la a0, array
16  li t0, 1
17
18  for_loop:
19  bge t0, a1, end      # branch hazard and dependency from line 39
20  slli t1, t0, 2
21  add t1, a0, t1       # add rs2 depends on rd of slli
22  lw t2, 0(t1)
23  addi t3, t0, -1
24
25  while_loop:
26  blt t3, x0, insert   # branch hazard and dependency from line 32
27  slli t4, t3, 2
28  add t4, a0, t4       # add rs2 depends on rd of slli
29  lw t5, 0(t4)         # here there is a load use hazard where lw writes t5
30  ble t5, t2, insert   # in MEM/WB, but ble needs t5 in EX
31  sw t5, 4(t4)         # similarly, t5 written in MEM/WB but need as input for sw
32  addi t3, t3, -1
33  beq x0, x0, while_loop
34
35  insert:
36  slli t4, t3, 2
37  add t4, a0, t4       # add rs2 depends on rd of slli
38  sw t2, 4(t4)
39  addi t0, t0, 1
40  beq x0, x0, for_loop # branch hazard
41
42  end:
43  li a7, 10
44  ecall
```